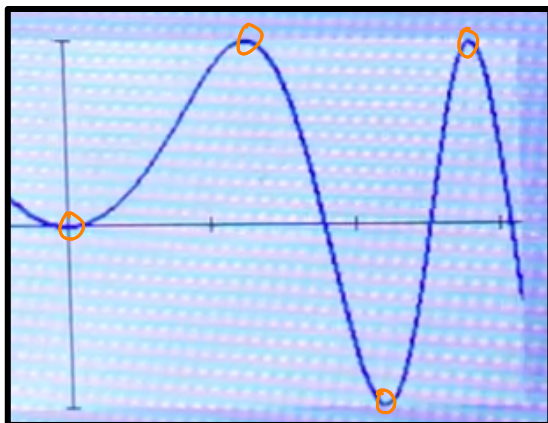


Section 3.4 The Chain Rule

30) Find all values in $[0, \pi)$ so that $\frac{dy}{dx} = 0$ for the function $y = \sin(x^2)$.

using a calculator we see visually there are 4 places where the derivative equals zero. (no slope)

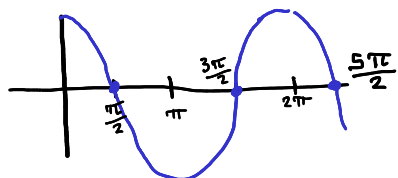


$$y' = \underbrace{\cos(x^2)}_{\text{set equal to zero}} \cdot \underbrace{2x}_{\text{set equal to zero}} = 0 \quad (\text{set equal to zero})$$

$$\cos(x^2) = 0$$

$$2x = 0$$

$$x = 0$$



$$x^2 = \frac{\pi}{2}$$

$$x^2 = \frac{3\pi}{2}$$

$$x^2 = \frac{5\pi}{2}$$

$$x = \sqrt{\frac{\pi}{2}}$$

$$x = \sqrt{\frac{3\pi}{2}}$$

$$x = \sqrt{\frac{5\pi}{2}}$$

$$x = 0, \sqrt{\frac{\pi}{2}}, \sqrt{\frac{3\pi}{2}}, \sqrt{\frac{5\pi}{2}}$$

31) Find $f'(x)$ for the function $f(x) = \sqrt{\frac{x}{x^2+1}}$.

- Derivative of Square root. $\rightarrow$
- Chain Rule and Quotient Rule.

$$\frac{d}{dx} (\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$f'(x) = \frac{1}{2\sqrt{\frac{x}{x^2+1}}} \cdot \left(\frac{1(x^2+1) - x \cdot 2x}{(x^2+1)^2} \right)$$

$$f'(x) = \frac{-x^2+1}{2 \frac{\sqrt{x}}{\sqrt{x^2+1}} \cdot (x^2+1)^2} \cdot \frac{(\sqrt{x^2+1})}{(\sqrt{x^2+1})}$$

$$f'(x) = \frac{(-x^2+1)(\sqrt{x^2+1})}{2\sqrt{x} (x^2+1)^2}$$

32) Find $f'(t)$ for the function $f(t) = \sin^2(\cos e^t)$.

Chain Rule


$$f(t) = \left(\sin(\cos(e^t)) \right)^2$$

$$f'(t) = 2(\sin(\cos(e^t))) \cdot \cos(\cos(e^t)) \cdot (-\sin(e^t)) \cdot e^t$$

$$f'(t) = -2(\sin(\cos(e^t))) \cdot \cos(\cos(e^t)) \cdot \sin(e^t) \cdot e^t$$

Section 3.5 Implicit Differentiation

33) Find $\frac{dy}{dx}$ given that $y \tan x - x^3 y^3 = \sin y$.

$$\underline{y'} \cdot \tan x + y \cdot \sec^2 x - (3x^2 \cdot y^3 + \underline{x^3 \cdot 3y^2 \cdot y'}) = \underline{\cos y \cdot y'}$$

$$y' \cdot \tan x - \cos y \cdot y' - x^3 \cdot 3y^2 \cdot y' = 3x^2 \cdot y^3 - y \cdot \sec^2 x$$

$$y' \cdot [\tan x - \cos y - x^3 \cdot 3y^2] = 3x^2 \cdot y^3 - y \cdot \sec^2 x$$

$$y' = \frac{3x^2 y^3 - y \sec^2 x}{\tan x - \cos y - x^3 3y^2}$$

34) Find $\frac{dy}{dx}$ for these three functions:

a) $x = \tan^{-1}(y^2)$

b) $x = \tan^{-1}(y)^2$
 $(\tan^{-1} y)^2$

c) $x = \tan^2(y^{-1})$

a) $1 = \frac{1}{1 + (y^2)^2} \cdot 2y \cdot y'$

$$\frac{1 + y^4}{2y} = y'$$

b) $1 = 2(\tan^{-1} y) \cdot \frac{1}{1 + y^2} \cdot y'$

$$\frac{1 + y^2}{2 \tan^{-1} y} = y'$$

c) $x = \left(\tan \frac{1}{y}\right)^2$
 $1 = 2\left(\tan \frac{1}{y}\right) \cdot \sec^2\left(\frac{1}{y}\right) \cdot \left(-\frac{1}{y^2}\right) \cdot y'$

$$\frac{-y^2}{2 \tan\left(\frac{1}{y}\right) \sec^2\left(\frac{1}{y}\right)} = y'$$

Section 3.6 Derivatives of Logarithmic Functions

35) Find $f'(x)$ for the function $f(x) = \ln\left(\frac{1}{x}\right) + \frac{1}{\ln x} + \ln(\ln x)$.

$$f'(x) = \frac{1}{\frac{1}{x}} - \frac{1}{(\ln x)^2} \cdot \frac{1}{x} + \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$f'(x) = x - \frac{1}{x \cdot (\ln x)^2} + \frac{1}{x \cdot \ln x}$$

36) Use logarithmic differentiation to find the first derivative of $y = (\cos x)^{\sin x}$.

$$\ln y = \ln(\cos x)^{\sin x}$$

$$\ln y = \sin x \cdot \ln(\cos x)$$

$$\frac{1}{y} \cdot y' = \cos x \cdot \ln(\cos x) + \sin x \cdot \frac{1}{\cos x} \cdot (-\sin x)$$

$$y' = y \left[\cos x \cdot \ln(\cos x) - \frac{\sin^2 x}{\cos x} \right]$$

$$y' = (\cos x)^{\sin x} \left[\cos x \cdot \ln(\cos x) - \frac{\sin^2 x}{\cos x} \right]$$

Section 3.7 Rates of Change

37) The population of the US is modeled by the function $P(t) = 309674727e^{0.021t}$ where $t = 0$ represents July 1, 2010. Find and explain the meaning of $P'(10.5)$.

$$P'(t) = 309674727 e^{0.021t} \cdot (0.021)$$

$$P'(10.5) = 309674727 e^{0.021(10.5)} (0.021)$$

$$= 8,107,500.636$$

In 2021 the us. population

will be growing by

8,107,500.636 people per year

38) Volume of a sphere is given by $V = \frac{4}{3}\pi r^3$. Suppose that the radius of a spherical balloon is shrinking at a rate of 2 in/min . Find and explain what $\frac{dV}{dt}$ is when $r = 8 \text{ in}$.

$$\frac{dr}{dt} = -2$$

$$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = \frac{4}{\cancel{3}}\pi \cdot \cancel{3}(8)^2 \cdot (-2)$$

$$\frac{dV}{dt} = 4\pi \cdot 64 \cdot (-2)$$

$$\frac{dV}{dt} = -1608.495 \frac{\text{in}^3}{\text{min}}$$

Volume is decreasing at
 $1608.495 \frac{\text{in}^3}{\text{min}}$ when $r = 8''$